

MARLBORO COUNTY AP, SC, USA

WMO: 720631

Lat: 34.617N

Lon: 79.733W

Elev: 45

StdP: 100.78

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00217

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.0	(c) -2.3	(d) -14.9	(e) 1.0	(f) -1.1	(g) -11.3	(h) 1.4	(i) 1.9	(j) 9.3	(k) 14.0	(l) 8.4	(m) 11.2	(n) 1.4	(o) 310	(p) 0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.3	(c) 36.3	(d) 23.9	(e) 35.1	(f) 23.6	(g) 33.6	(h) 23.2	(i) 26.6	(j) 31.2	(k) 26.0	(l) 30.6	(m) 25.5	(n) 30.1	(o) 3.0	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.3	20.5	27.5	25.1	20.3	27.4	24.8	19.9	27.3	83.5	30.7	80.7	30.3	78.1	30.1	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.7	6.7	5.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			-8.5	27.9	3.9	0.7	-11.3	28.3	-13.6	28.7	-15.7	29.1	-18.6	29.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	7.5	9.1	12.4	17.8	22.7	26.9	27.6	26.5	24.0	17.9	11.6	9.3
	DBStd	8.37	5.46	5.24	5.08	4.26	3.43	2.37	2.09	2.58	2.88	4.40	4.36	5.08
	HDD10.0	329	112	74	32	2	0	0	0	0	2	31	77	
	HDD18.3	1422	336	263	196	62	8	0	0	0	2	62	208	285
	CDD10.0	3183	36	49	106	235	393	508	547	511	421	247	78	54
	CDD18.3	1234	1	4	12	45	143	258	288	253	172	49	5	4
	CDH23.3	12341	7	32	121	501	1335	2716	3044	2530	1537	469	35	13
	CDH26.7	5484	0	2	18	138	546	1354	1487	1171	637	128	2	0
(14) Wind	WSAvg	2.0	2.2	2.5	2.6	2.5	2.1	2.0	1.9	1.5	1.7	1.8	1.8	2.0
(15) Precipitation	PrecAvg	1145	86	82	93	76	73	118	123	130	121	88	76	76
	PrecMax	1375	183	167	156	156	145	241	231	241	273	269	207	153
	PrecMin	782	24	36	19	22	36	20	35	29	25	0	11	25
	PrecStd	139	45	33	36	34	29	57	53	55	69	61	48	36
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.9	27.2	28.0	32.2	35.1	37.7	37.7	37.2	35.1	32.4	27.0	25.1
		MCWB	18.0	18.4	17.6	19.7	21.3	24.3	24.3	24.3	22.3	23.7	19.0	20.6
	2%	DB	21.4	23.7	27.1	30.1	32.8	36.2	36.2	35.2	32.9	30.0	23.9	22.6
		MCWB	16.5	17.5	17.6	19.2	20.6	23.4	24.5	24.4	23.3	21.5	17.2	19.3
	5%	DB	18.9	21.1	24.7	27.8	32.1	34.9	35.0	33.6	32.2	27.6	22.3	21.0
		MCWB	15.3	16.2	16.4	18.6	20.5	23.0	24.2	24.2	23.2	20.3	16.6	18.7
	10%	DB	17.0	17.9	22.1	26.3	30.1	32.8	32.8	32.4	31.0	26.1	20.1	17.9
		MCWB	13.3	14.1	15.2	17.8	20.7	22.7	23.9	24.0	23.0	20.1	15.8	15.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.0	20.5	20.3	23.5	24.9	26.8	27.4	27.3	26.8	25.8	21.6	22.2
		MCDB	22.0	23.1	24.3	27.7	27.3	32.6	33.6	31.5	29.2	27.6	23.0	24.2
	2%	WB	18.1	19.0	18.9	21.9	23.8	25.7	26.4	26.7	25.7	24.8	20.1	20.9
		MCDB	20.5	22.7	23.9	25.6	27.5	31.6	31.9	30.9	28.8	27.7	22.2	21.8
	5%	WB	16.2	17.6	17.8	20.6	22.9	25.2	25.8	26.0	25.0	22.9	18.4	18.6
		MCDB	17.8	19.9	21.8	25.0	26.8	30.8	31.0	30.4	28.6	26.0	21.1	20.4
	10%	WB	13.5	15.1	16.7	19.3	22.3	24.6	25.3	25.5	24.2	21.3	16.7	16.1
		MCDB	15.8	18.1	20.8	24.2	26.7	30.1	30.4	29.9	28.7	23.6	19.2	18.2
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	11.6	12.8	13.9	12.5	12.0	11.3	11.2	12.0	13.3	13.4	11.1
		MCDBR	14.7	14.3	15.4	15.6	15.2	13.8	13.0	13.3	14.2	15.4	15.6	12.5
		MCWBR	9.5	8.8	7.4	6.7	5.7	4.3	4.0	4.7	5.6	8.0	9.3	8.7
	5% WB	MCDBR	11.4	12.9	12.7	12.1	11.2	11.6	11.0	11.1	11.0	11.0	11.7	11.6
		MCWBR	9.3	8.9	7.4	6.3	4.9	4.3	4.1	4.9	5.0	6.8	8.7	9.1
(40) Clear-Sky Solar Irradiance	taub		0.317	0.327	0.363	0.405	0.468	0.493	0.543	0.515	0.431	0.391	0.339	0.314
	taud		2.499	2.466	2.398	2.268	2.147	2.118	2.000	2.068	2.294	2.371	2.475	2.545
	Ebn at Noon		892	920	910	881	827	803	760	774	829	835	856	875
	Edh at Noon		84	96	112	134	153	158	176	162	123	105	86	76
(44) All-Sky Solar Radiation	RadAvg		2.71	3.35	4.44	5.57	6.09	6.39	6.07	5.51	4.71	3.88	2.97	2.36
	RadStd		0.15	0.36	0.34	0.39	0.49	0.43	0.38	0.26	0.40	0.47	0.24	0.20

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+123

Nomenclature: See separate page